STOCK MARKET ANALYSIS AND PREDICITON

A Data-Driven Approach to Forecasting Financial Markets

PRESENTED BY - NEETU KUMARI & SHUBHAM GUPTA

CAPSTONE PROJECT - DATA SCIENCE & AI

IIT Roorkee

WHY STOCK PRICE PREDICTION MATTERS

- Stock markets are volatile and influenced by complex factors like news, sentiment, and macroeconomic indicators.
- Accurate prediction of stock prices helps investors make better decisions and reduce financial risks.
- This project aims to understand past trends and forecast future prices using data science techniques.
- Combines statistical methods and machine learning models to analyze stock performance over time.
- Moving averages and volume dynamics are used to uncover potential buy/sell windows.



WHAT WE SET OUT TO DISCOVER

- Can we analyze and forecast the price movements of top tech stocks using historical data?
- How do market indicators like closing price, volume, and moving averages reflect stock behavior over time?
- Is it possible to extract actionable insights for investors by studying these patterns?
- We focused on four tech leaders: Apple (AAPL), Google (GOOG), Microsoft (MSFT), and Amazon (AMZN).
- Our goal: Combine financial data with data science techniques to gain clarity on price trends and their precision.

DATA COLLECTION AND DATASET OVERVIEW

- **Source**: Historical stock price data retrieved from Yahoo Finance API using *yfinance* for stocks Amazon, Apple, Google and Microsoft.

- **Timeframe**: Past 1 year (customizable depending upon when run)

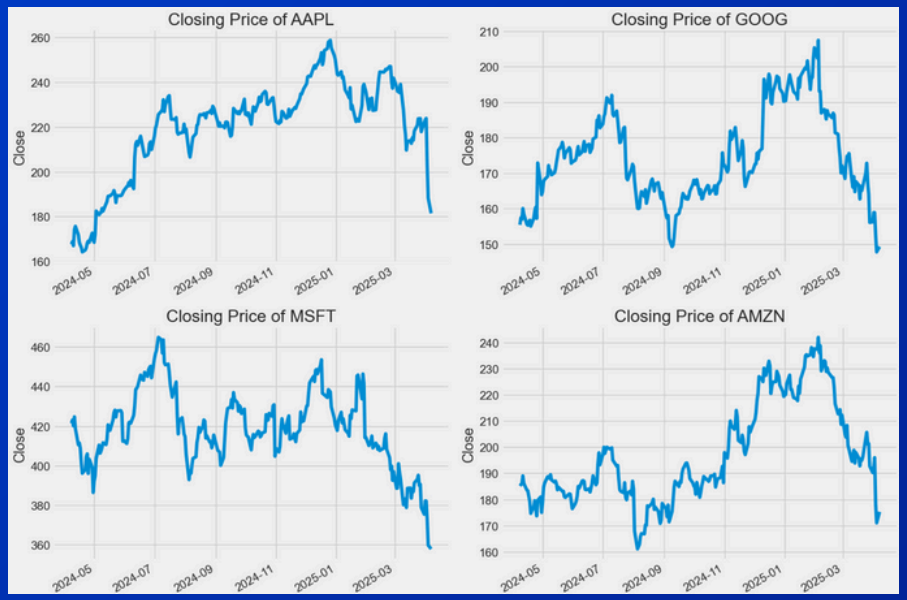
- **Key fields analyzed:**
  Date, Open, High, Low, Close, Adj Close, Volume

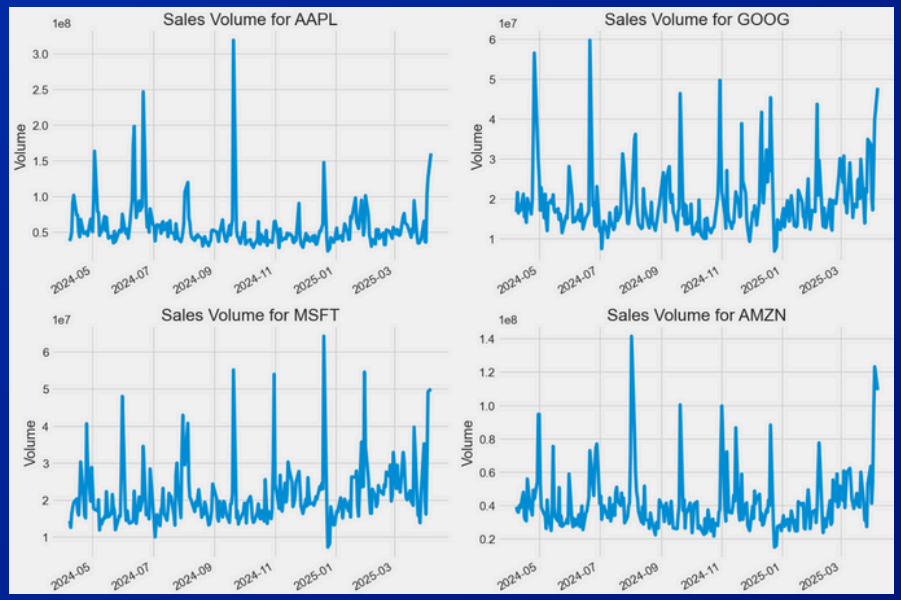
- **Data Format**: CSV with daily frequency

- **Data Quality**: Data was clean and structured — ready for visualization and trend analysis.

Price Date	Close	High	Low	Open	Volume	Company
2025-03-25	205.710007	206.210007	203.220001	203.600006	31171200	Amazon
2025-03-26	201.130005	206.009995	199.929993	205.839996	32855300	Amazon
2025-03-27	201.360001	203.789993	199.279999	200.889999	27317700	Amazon
2025-03-28	192.720001	199.259995	191.880005	198.419998	52548200	Amazon
2025-03-31	190.259995	191.330002	184.399994	188.190002	63547600	Amazon
2025-04-01	192.169998	193.929993	187.199997	187.860001	41267300	Amazon

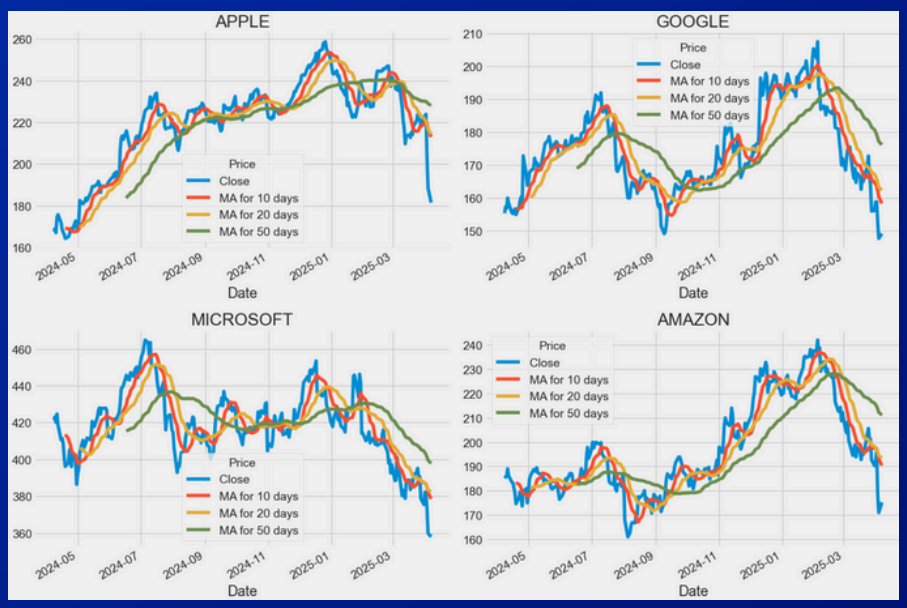
TREND ANALYSIS AND MARKET BEHAVIOR





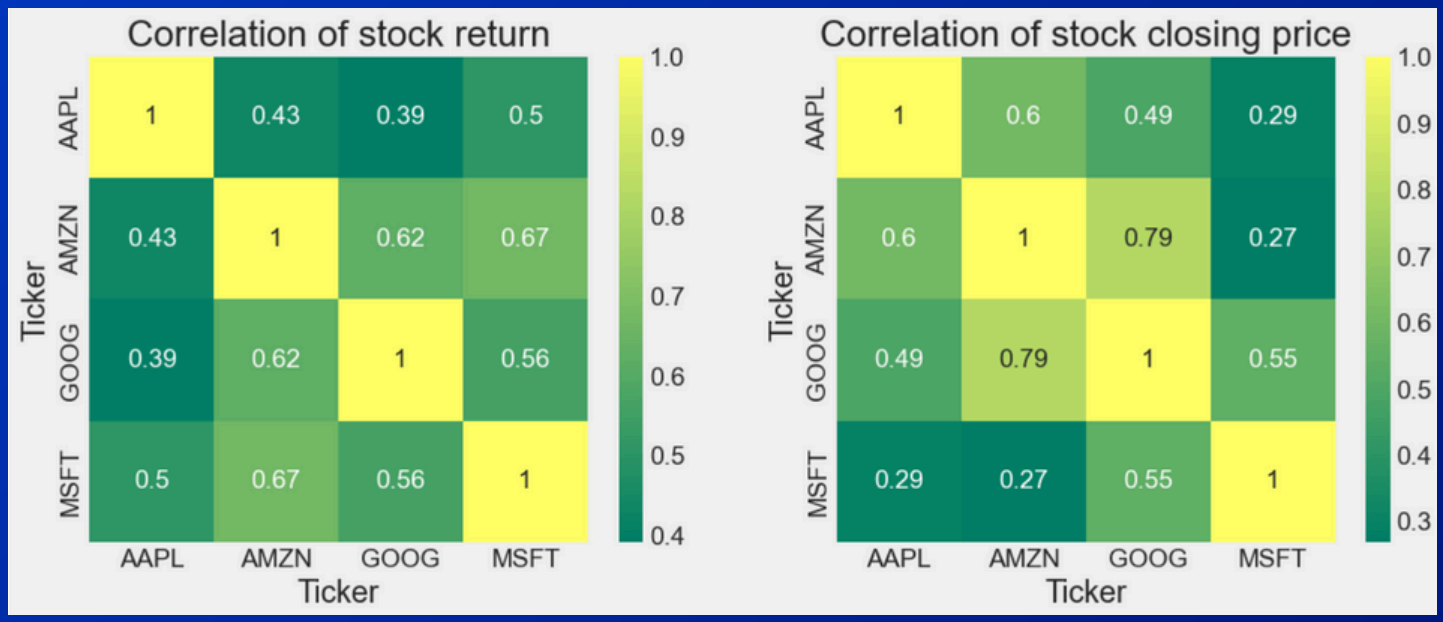
- **Closing Price Patterns:** The most stable (least jagged) line is for AAPL; the most volatile appears to be GOOG and AMZN. The recent downtrend across all 4 might reflect a broader market correction or industry shift.

- **Volume Movement:** AAPL and AMZN show much higher peaks, suggesting that investors react more intensely to these companies' news.

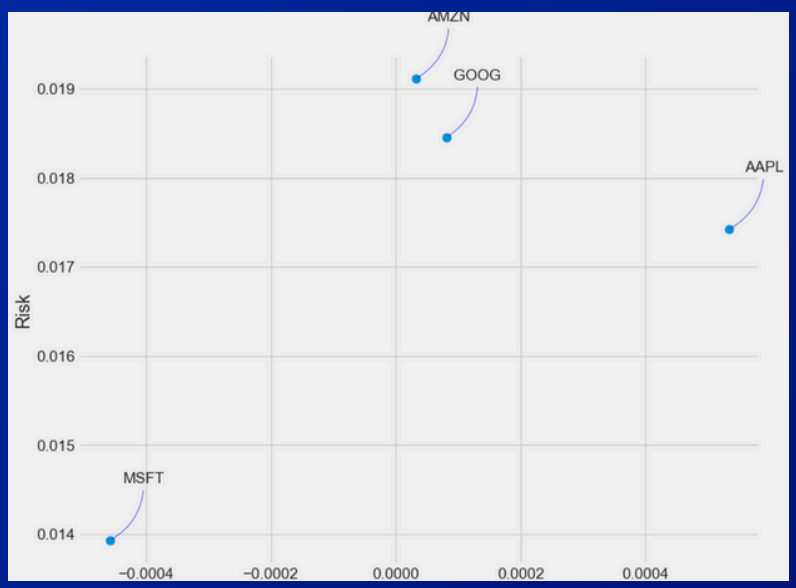


- **Moving Average Insights:** All four companies are showing signs of trouble lately. The recent drop in prices for all of them suggests that people may be starting to sell their shares

TREND ANALYSIS AND MARKET BEHAVIOR - CONTINUED

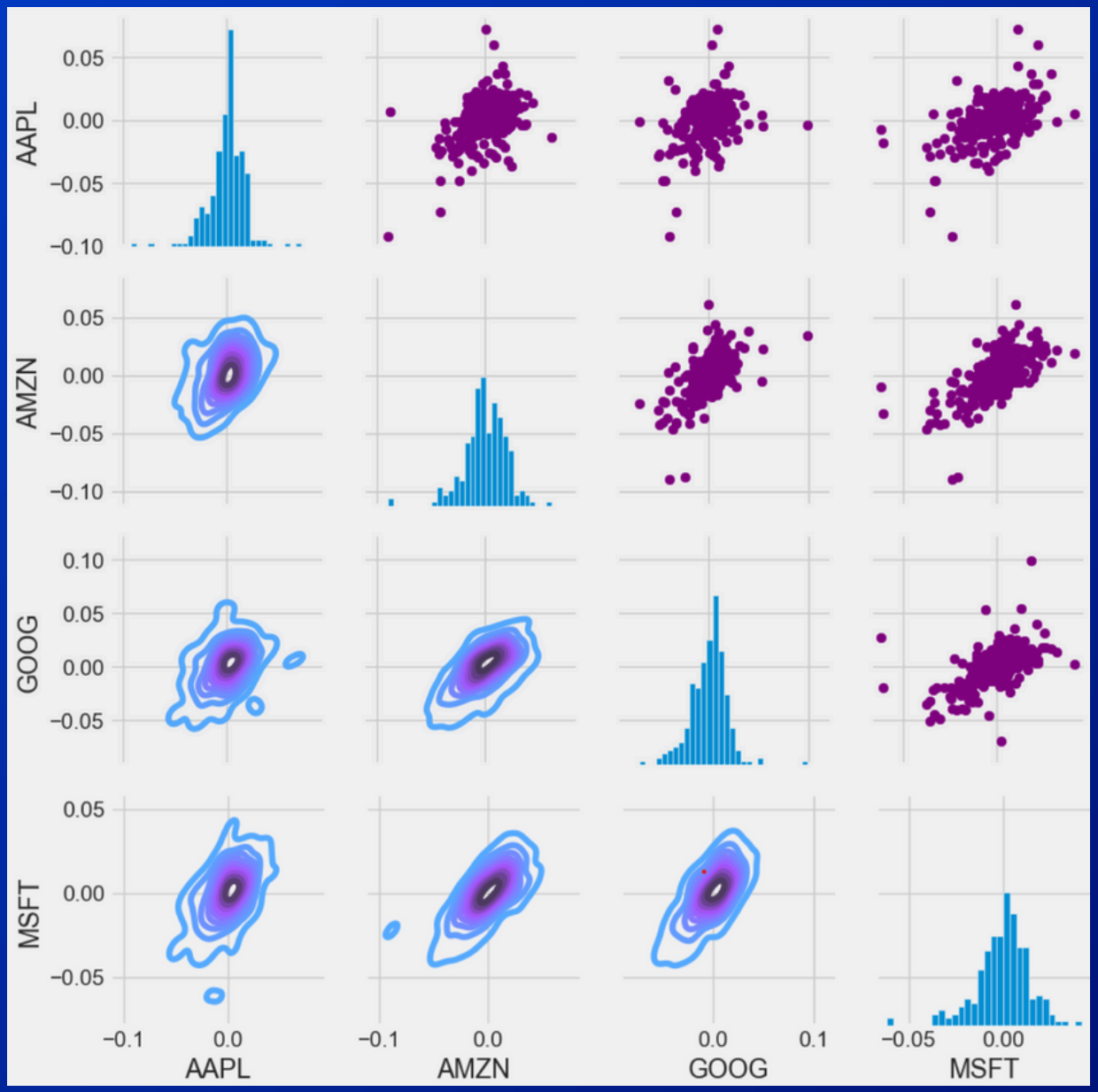


- **Correlation of Stock :** All values are between 0.39 and 0.67, indicating a moderate positive correlation in returns. MSFT and AMZN have the highest correlation in returns (0.67) — they tend to move similarly day to day. AAPL and GOOG have the lowest correlation in returns (0.39), meaning they move more independently. Returns are more volatile and less tightly linked.



- **Risk v/s Return :** This graph of risk versus expected return shows that if we are looking for safety, Microsoft is the least risky—but you might not earn much (or could even lose a bit). If we want the best chance to earn more, Apple looks like the most promising—but with slightly more risk.

FEATURE ENGINEERING



- **What Was Done:** In this project, daily stock returns were calculated using the percentage change in closing prices.

- **Observations** :

1. Return distributions were centered around 0%, as expected for daily market data.
2. Some stocks (like Amazon) showed wider spreads, indicating higher volatility.
3. Microsoft's returns were more tightly grouped, showing more stability.

ALGORITHM AND MODEL USED

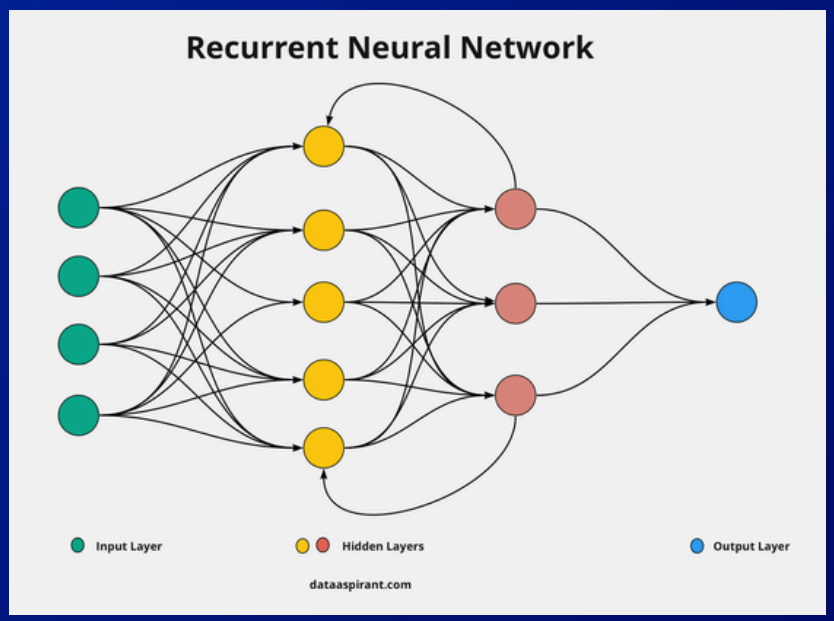
- **Algorithm Used: LSTM** (Long Short-Term Memory) — a type of Recurrent Neural Network (RNN)

- **What is LSTM?** : LSTM is a deep learning algorithm designed to process and make predictions on time-series data, such as stock prices.

1. Captures temporal dependencies — learns from patterns over time
2. Handles long-term trends better than standard RNNs
3. Well-suited for sequential data like financial market behavior.

- **What it shows**: A simple pipeline showing:
  - Input data → LSTM layer → Dense layer → Output (prediction)

- **Why it works**: Represents your actual model structure in a clean, abstract form.



MODEL EVALUATION AND RESULTS



Price	Close	Predictions
Date		
2024-08-08	212.595474	208.048096
2024-08-09	215.515686	207.401428
2024-08-12	217.052292	208.342789
2024-08-13	220.784073	210.150604
2024-08-14	221.233093	212.754044
...	...	...
2025-04-01	223.190002	218.013092
2025-04-02	223.889999	217.959686
2025-04-03	203.190002	218.276871
2025-04-04	188.380005	214.092789
2025-04-07	181.460007	205.848480

- **Prediction Accuracy** (Numerical Check): For example, on April 7, 2025, the actual close was USD 181.46, but the model predicted USD 205.85 — an overestimation of over USD 24.

- **Trend Overestimation**: In the last week of data, actual prices dropped sharply, but predictions remained somewhat higher. This again shows the model was expecting the trend to continue upward or stabilize, and didn't anticipate a sharp decline.

- **General Trend Matching**: Despite some misses, most predictions earlier on (e.g., in August 2024) are fairly close to the actual values — often within USD 5, which is decent.

MODEL CONCLUSION

- The model performs well during stable or upward trends, but is less responsive to sudden market shocks or trend reversals.

- In recent data, Apple’s stock had a significant drop, which the model did not fully capture — indicating possible news or events that the model, trained only on past prices, couldn't predict.

- The predictions are reasonably accurate and useful for short-term forecasting in normal market conditions, but should not be relied on alone for critical investment decisions.

- This kind of model is a good helper tool, but real-world decisions should also consider news, macroeconomic factors, and expert analysis.

THANK YOU